

Deep e-prop in a deep leaky RNN — forward eligibility-trace recursion

trace ϵ_t^l per parameter group, propagated forward in time *and* up in depth ($l = 1, \dots, L$)

$$\epsilon_t^l = \underbrace{\frac{\partial h_t^l}{\partial h_{t-1}^l}}_{\text{recurrent (in time)}} \epsilon_{t-1}^l + \begin{cases} \frac{\partial h_t^l}{\partial h_t^{l-1}} \epsilon_t^{l-1}, & 1 < l \leq L \quad \text{hierarchical (in depth)} \\ \frac{\partial h_t^l}{\partial \theta}, & l = 1 \quad \text{input source (terminal)} \end{cases}$$

with the leaky-integrator Jacobians ($\psi_t^l = \tanh'(a_t^l) = 1 - \tanh^2(a_t^l)$, $a_t^l = W_{\text{rec}}^l h_{t-1}^l + W_{\text{ff}}^l h_t^{l-1} + b^l$):

$$\frac{\partial h_t^l}{\partial h_{t-1}^l} = \underbrace{(1 - \alpha) I}_{\text{leak / carry}} + \alpha \text{diag}(\psi_t^l) W_{\text{rec}}^l \qquad \frac{\partial h_t^l}{\partial h_t^{l-1}} = \alpha \text{diag}(\psi_t^l) W_{\text{ff}}^l$$

loss gradient (loss at any / every step), via the non-recurrent learning signal on the top layer L :

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_t \frac{\partial \mathcal{L}_t}{\partial y_t} \frac{\partial y_t}{\partial h_t^L} \epsilon_t^L$$

Here ∂ (rather than the total d) is the e-prop approximation: indirect cross-time parameter interactions are dropped, so each trace stays local and online — one per synapse, per layer.